RTEL1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RTEL1 gene fusions from the msk_impact_50k_2026 study, covering 24 fusion events. 5/10 fusions with known frame status (50.0%) were in-frame. 9/24 fusions (37.5%) retained the Helicase C-terminal domain. Domain retention was tested with Fisher's exact test (p=0.116746, not statistically significant at alpha=0.05). No literature benchmark is configured for RTEL1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 24 analyzed fusion events, identifying 19 distinct fusion partner genes. The most frequent partner was ZGPAT, observed in 4 events.

Domain retention

Helicase C-terminal domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/5 (60.0%) of in-frame fusions retained the domain, p=0.116746 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.0609391.

Domain disruption

Disruption of the DEAD_2 was tested with Fisher's exact test comparing in-frame fusions against all others; 1/4 (25.0%) of in-frame fusions disrupted the domain, p=0.990431 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.970297.

Junction concentration

17% of 12 fusions with this gene as the 3' partner join at exon 26, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.0599401, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 22 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 357 aa (exon 13; label-permutation p=0.653347). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 85 aa from the nearest configured domain boundary at 272 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=8) and "not_retained" (n=14). For Frame_status, retained: 3/4 (75.0%, 95% CI 30.1%-95.4%); not_retained: 1/5 (20.0%, 95% CI 3.6%-62.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 32.25 vs 40.2857, p=0.696709 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for RTEL1 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (13 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: RTEL1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: RTEL1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (38 row(s) in window_scan, 22 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZGPAT	4	0.16666666666666666	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.13945753425043486	1
CDH4	2	0.08333333333333333	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11004576954455252	2
TSHZ2	2	0.08333333333333333	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11004576954455252	3
ARFGAP1	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	4
ARFRP1	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	5
BIRC7	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	6
CPQ	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	7
DNAJC5	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	8
PKP2	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	9
PRPF6	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	10
PSMA7	1	0.041666666666666664	0.12151042424685517	0.0013095298090147 73	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RIN2	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	12
SEC23B	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	13
SPTLC3	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	14
STAU1	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	15
STMN3	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	16
TAF4	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	17
VRK3	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	18
YBX2	1	0.041666666666666664	0.12151042424685517	0.001309529809014773	None	0.3789939208935289	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09533988719161132	19

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
35	1	7	0	14	None	0.36363636363636365	0.43933269383026263
68	1	7	1	13	1.8571428571428572	1.0	-0.0
159	1	7	2	12	0.8571428571428571	1.0	-0.0
176	1	7	3	11	0.5238095238095238	1.0	-0.0
307	1	7	4	10	0.35714285714285715	0.6129718234981393	0.21255948825567986
346	1	7	5	9	0.2571428571428571	0.3511586452762923	0.4544966350945685
357	1	7	6	8	0.19047619047619047	0.19321699971854772	0.713954665916554
423	2	6	6	8	0.4444444444444444	0.6493980048159614	0.18748905001668414

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
446	3	5	6	8	0.8	1.0	-0.0
457	4	4	6	8	1.3333333333333333	1.0	-0.0
604	4	4	7	7	1.0	1.0	-0.0
636	4	4	8	6	0.75	1.0	-0.0
744	4	4	9	5	0.5555555555555556	0.6619195046439628	0.17919482145834373
756	4	4	11	3	0.2727272727272727	0.3426212590299278	0.465185693346729
884	5	3	11	3	0.45454545454545453	0.6243550051599589	0.20456840239829632
998	6	2	11	3	0.8181818181818182	1.0	-0.0
1006	6	2	12	2	0.5	0.6019138755980862	0.22046564500180385
1037	8	0	12	2	None	0.5151515151515151	0.2880650184996136
1085	8	0	13	1	None	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

1	13
3	5

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[0, 0], [0, 1]
BLCA	[0, 0], [1, 0]
BRCA	[1, 0], [0, 0]
CCOV	[0, 1], [0, 0]
COAD	[0, 0], [0, 1]
CUP	[0, 1], [0, 0]
ESCA	[0, 0], [1, 0]
GEJ	[0, 1], [0, 0]
HGSOC	[0, 3], [0, 0]
IDC	[0, 1], [0, 1]
ILC	[0, 0], [0, 1]
LUAD	[0, 2], [1, 1]
LUCA	[0, 1], [0, 0]
MFH	[0, 1], [0, 0]

stratum	table
PAAD	[0, 1], [0, 0]
PDC	[0, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf06733	DEAD_2	22	8	2	12	0.05070546737213404

domain_retention tables

frame_domain_contingency_table

3	5
1	13

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[0, 1], [0, 0]
BLCA	[1, 0], [0, 0]
BRCA	[1, 0], [0, 0]
CCOV	[0, 0], [0, 1]
COAD	[0, 1], [0, 0]
CUP	[0, 0], [0, 1]
ESCA	[1, 0], [0, 0]
GEJ	[0, 0], [0, 1]
HGSOC	[0, 1], [0, 2]
IDC	[0, 0], [0, 2]
ILC	[0, 0], [0, 1]
LUAD	[0, 1], [1, 2]
LUCA	[0, 1], [0, 0]
MFH	[0, 0], [0, 1]
PAAD	[0, 0], [0, 1]
PDC	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Helicase C-terminal domain	22	8	2	12	0.08602150537634409

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
19	8	22	0.36363636363636365

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ARFGAP1	1	1	1	1
ARFRP1	1	1	1	1
BIRC7	1	1	1	1
CDH4	2	2	2	1
CPQ	1	1	1	1
DNAJC5	1	1	1	1
PKP2	1	1	1	1
PRPF6	1	1	1	1
PSMA7	1	1	1	1
RIN2	1	1	1	1
SEC23B	1	1	1	1
SPTLC3	1	1	1	1
STAU1	1	1	1	1
STMN3	1	1	1	1
TAF4	1	1	1	1
TSHZ2	2	2	2	1
VRK3	1	1	1	1
YBX2	1	1	1	1
ZGPAT	4	4	4	4

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
20	62325000	62326000	4	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
20	62321000	62322000	3	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0053173-T01-IM6-2
20	62292000	62293000	2	EVT-P-0035723-T01-IM6-47, EVT-P-0041523-T01-IM6-9
20	62294000	62295000	2	EVT-P-0019862-T01-IM6-51, EVT-P-0052166-T01-IM6-6
20	62320000	62321000	2	EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57
20	62326000	62327000	2	EVT-P-0014391-T01-IM6-50, EVT-P-0065831-T01-IM7-1
20	62305000	62306000	1	EVT-P-0044707-T03-IM6-8
20	62309000	62310000	1	EVT-P-0057597-T01-IM6-16
20	62311000	62312000	1	EVT-P-0045154-T01-IM6-43
20	62316000	62317000	1	EVT-P-0004381-T02-IM6-3
20	62317000	62318000	1	EVT-P-0026247-T02-IM6-48
20	62319000	62320000	1	EVT-P-0028716-T01-IM6-52
20	62323000	62324000	1	EVT-P-0022005-T01-IM6-10

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
756	2	2	68	False
1037	2	2	3	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	31	2	10	0.2	0.009725370302039956	2.39350990061384
three_prime	16	2	12	0.16666666666666666	0.007247904696642243	2.481009073795158
three_prime	26	2	12	0.16666666666666666	0.006088239945179485	2.6290950137092026
five_prime	3	1	10	0.1	0.014548521427441886	0.8654378354105319
five_prime	5	1	10	0.1	0.007458752833271836	1.1418648385837353

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	6	1	10	0.1	0.08368035422486954	0.23457044717191922
five_prime	10	1	10	0.1	0.11082705181592958	0.1604790013307032
five_prime	12	1	10	0.1	0.04166886300142322	0.46012782143532777
five_prime	13	1	10	0.1	0.02148015391913974	0.7095548854157963
five_prime	28	1	10	0.1	0.028016446154657107	0.6066762196880026
five_prime	30	1	10	0.1	0.032391544989721154	0.5519792873610784
three_prime	3	1	12	0.08333333333333333	0.052501186020768545	0.32196721523687205
three_prime	17	1	12	0.08333333333333333	0.05002372041537083	0.3374293365085039
three_prime	22	1	12	0.08333333333333333	0.013889621000474408	0.8110318228757339
three_prime	23	1	12	0.08333333333333333	0.013599704812608719	0.8195116315491785
three_prime	25	1	12	0.08333333333333333	0.009092825892151178	0.9837226415002485
three_prime	31	1	12	0.08333333333333333	0.03175900057983238	0.4933508442237222
three_prime	32	1	12	0.08333333333333333	0.012809024300247747	0.8436662542294603
three_prime	33	1	12	0.08333333333333333	0.00650993621843867	1.122732240568756

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-24	176	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0019862-T01-IM6-51, EVT-P-0035723-T01-IM6-47, EVT-P-0041523-T01-IM6-9, EVT-P-0052166-T01-IM6-6
35	235	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0019862-T01-IM6-51, EVT-P-0035723-T01-IM6-47, EVT-P-0041523-T01-IM6-9, EVT-P-0052166-T01-IM6-6
146	346	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0019862-T01-IM6-51, EVT-P-0044707-T03-IM6-8, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
157	357	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0019862-T01-IM6-51, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
159	359	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0019862-T01-IM6-51, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
176	376	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
223	423	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0004381-T02-IM6-3, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
246	446	200	5	17	2	6	1.2222222222222223	1.0	-0.0	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
257	457	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, ... (6 total)
307	507	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, ... (6 total)
346	446	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
346	546	200	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
357	457	100	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0045154-T01-IM6-43
357	557	200	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0045154-T01-IM6-43
404	604	200	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55
423	623	200	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55
436	636	200	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
446	646	200	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57
556	756	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57, EVT-P-0053173-T01-IM6-2
604	804	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57, EVT-P-0053173-T01-IM6-2
636	836	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0049269-T01-IM6-57, EVT-P-0053173-T01-IM6-2
684	884	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0022005-T01-IM6-10, EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0053173-T01-IM6-2
744	944	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0022005-T01-IM6-10, EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0053173-T01-IM6-2
837	1037	200	5	17	4	4	13.0	0.03934077618288144	1.4051570758194027	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0022005-T01-IM6-10, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
884	1084	200	5	17	4	4	13.0	0.03934077618288144	1.4051570758194027	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0022005-T01-IM6-10, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
885	1085	200	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
930	1130	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44, ... (6 total)
937	1037	100	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
985	1085	100	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
987	1037	50	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
998	1048	50	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
998	1098	100	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
998	1198	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44, ... (6 total)
1006	1106	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46
1006	1206	200	5	17	2	6	1.2222222222222223	1.0	-0.0	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1
1030	1130	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1
1037	1137	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1
1037	1237	200	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
837	1037	200	5	17	4	4	13.0	0.03934077618288144	1.4051570758194027	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0022005-T01-IM6-10, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
157	357	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0019862-T01-IM6-51, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
556	756	200	5	17	0	8	0.0	0.11536416799574695	0.9379290617018493	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57, EVT-P-0053173-T01-IM6-2
987	1037	50	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
357	457	100	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0045154-T01-IM6-43
404	604	200	4	18	3	5	7.8	0.11674641148325358	0.9327564597723246	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55
146	346	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0019862-T01-IM6-51, EVT-P-0044707-T03-IM6-8, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
176	376	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0052166-T01-IM6-6, EVT-P-0057597-T01-IM6-16
636	836	200	4	18	0	8	0.0	0.2535885167464115	0.595870416510265	EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0049269-T01-IM6-57, EVT-P-0053173-T01-IM6-2
985	1085	100	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44
346	546	200	5	17	3	5	3.6	0.3088782562466773	0.5102126631600481	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
346	446	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1006	1106	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46
1030	1130	100	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1
436	636	200	4	18	2	6	2.0	0.6019138755980862	0.22046564500180385	EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0038164-T01-IM6-55, EVT-P-0049269-T01-IM6-57
257	457	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0028716-T01-IM6-52, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, ... (6 total)
930	1130	200	6	16	3	5	2.2	0.6243550051599587	0.20456840239829646	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0051006-T01-IM6-44, ... (6 total)
-24	176	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0019862-T01-IM6-51, EVT-P-0035723-T01-IM6-47, EVT-P-0041523-T01-IM6-9, EVT-P-0052166-T01-IM6-6
223	423	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0004381-T02-IM6-3, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
246	446	200	5	17	2	6	1.2222222222222223	1.0	-0.0	EVT-P-0004381-T02-IM6-3, EVT-P-0026247-T02-IM6-48, EVT-P-0044707-T03-IM6-8, EVT-P-0045154-T01-IM6-43, EVT-P-0057597-T01-IM6-16
684	884	200	4	18	1	7	0.5238095238095238	1.0	-0.0	EVT-P-0022005-T01-IM6-10, EVT-P-0024365-T01-IM6-4, EVT-P-0035764-T01-IM6-49, EVT-P-0053173-T01-IM6-2
1006	1206	200	5	17	2	6	1.2222222222222223	1.0	-0.0	EVT-P-0014391-T01-IM6-50, EVT-P-0017448-T01-IM6-54, EVT-P-0018660-T01-IM6-5, EVT-P-0047648-T01-IM6-46, EVT-P-0065831-T01-IM7-1

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006583 1-T01-IM7-1	P-006583 1-T01-IM7	P-006583 1	Non-Small Cell Lung Cancer	LUAD	ARFGAP1-RTEL1	ARFGAP1	unknown	three_prime	62326464	33	1130	False	1130	True	exact match	lost	0.0	False													Helicase C-terminal domain; DEAD_2		ARFGAP1 (NM_175609) - RTEL1 (NM_032957) rearrangement: c.717+287:ARFGAP1_c.3461:RTEL1del | Protein Fusion: mid-exon { ARFGAP1:RTEL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005317 3-T01-IM6-2	P-005317 3-T01-IM6	P-005317 3	Esophagogastric Cancer	GEJ	PRPF6-RTEL1	PRPF6	out-of-frame	three_prime	62321573	26	756	True	756	True	exact match	lost	0.0	False													Helicase C-terminal domain; DEAD_2		PRPF6 (NM_012469) - RTEL1 (NM_032957) rearrangement: c.360-1518:PRPF6_c.2337+10:RTEL1dup | Protein Fusion: out of frame {PRPF6:RTEL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00043 81-T02-IM6-3	P-00043 81-T02-IM6	P-00043 81	Breast Cancer	BRCA	PSMA7-RTEL1	PSMA7	in-frame	three_prime	62316995	16	423	True	423	True	exact match	retained	1.0	False													Helicase C-terminal domain	DEAD_2		PSMA7 (NM_002792) - RTEL1 (NM_032957) rearrangement: c.472-149:PSMA7_c.1338+45:RTEL1inv | Protein Fusion: in frame {PSMA7:RTEL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002436 5-T01-IM6-4	P-002436 5	P-002436 5	Ovarian Cancer	HGSOC	RIN2-RTEL1	RIN2	out-of-frame	three_prime	62321641	26	756	True	756	True	exact match	lost	0.0	False														Helicase C-terminal domain; DEAD_2		RIN2 (NM_001242581) -RTEL1 (NM_032957) Rearrangement : c.2348-1384:RIN2_c.2338-6:RTEL1 | Protein Fusion: out of frame {RIN2:RTEL1}	NA
EVT-P-001866 0-T01-IM6-5	P-001866 0	P-001866 0	Non-Small Cell Lung Cancer	LUAD	RTEL1-ARFRP1	ARFRP1	unknown	three_prime	62325750	31	1006	False	1006	True	exact match	lost	0.0	False														Helicase C-terminal domain; DEAD_2		RTEL1 (NM_032957) Rearrangement : c.3090:RTEL1_c.392:ARFRP1 del | Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005216-T01-IM6-6	P-005216-T01-IM6	P-005216	Ovarian Cancer	CCOV	RTEL1-BIRC7	BIRC7	unknown	five_prime	62294231	6	176	False	175	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Helicase C-terminal domain	DEAD_2	RTEL1 (NM_032957) - BIRC7 (NM_139317) rearrangement: c.599:RTEL1_c.649+16:BIRC7dup | Protein Fusion: mid-exon {RTEL1:BIRC7}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004470 7-T04-IM6-7	P-004470 7-T04-IM6	P-004470 7	Non-Small Cell Lung Cancer	LUAD	RTEL1-CDH4	CDH4	in-frame	five_prime	62305671	10	307	True	306	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False													DEAD_2	Helicase C-terminal domain			RTEL1 (NM_032957) - CDH4 (NM_001794) rearrangement: c.991+225:RTEL1_c.170-168444:CDH4dup | Protein Fusion: in frame {RTEL1:CDH4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004470 7-T03-IM6-8	P-004470 7-T03-IM6	P-004470 7	Non-Small Cell Lung Cancer	LUAD	RTEL1-CDH4	CDH4	in-frame	five_prime	62305671	10	307	True	306	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False													DEAD_2	Helicase C-terminal domain			RTEL1 (NM_032957) - CDH4 (NM_001794) rearrangement: c.991+225:RTEL1_c.170-168444:CDH4dup | Protein Fusion: in frame {RTEL1:CDH4}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004152 3-T01-IM6-9	P-004152 3-T01-IM6	P-004152 3	Breast Cancer	IDC	RTEL1-CPQ	CPQ	unknown	five_prime	62292750	3	68	False	67	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Helicase C-terminal domain; DEAD_2		RTEL1 (NM_032957)-CPQ (NM_016134.3) rearrangement: t(8;20)(q22.1;q13.33)(chr8:g.98069745::chr20:g.62292750)| Protein Fusion: mid-exon {RTEL1:CPQ}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002205-T01-IM6-10	P-002205-T01-IM6	P-002205	Cancer of Unknown Primary	ADNOS	RTEL1-DNAJC5	DNAJC5	unknown	five_prime	62323205	28	884	True	884	True	exact match	retained	1.0	False													Helicase C-terminal domain; DEAD_2			RTEL1 (NM_032957) rearrangement: c.2724+15:RTEL1_chr20:g.62528262del | Transcript Fusion {RTEL1:DNAJC5}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005597-T01-IM6-16	P-005597-T01-IM6	P-005597	Breast Cancer	ILC	RTEL1-PKP2	PKP2	out-of-frame	five_prime	62309782	12	346	True	345	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															DEAD_2	Helicase C-terminal domain			RTEL1 (NM_032957) - PKP2 (NM_001005242) rearrangement: t(12;20)(p11.21;q13.33)(chr12:g.33044173::chr20:g.62309782) | Protein Fusion: out of frame {RTEL1:PKP2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004514-T01-IM6-43	P-004514-T01-IM6	P-004514	Breast Cancer	IDC	RTEL1-SPTLC3	SPTLC3	unknown	five_prime	62311233	13	357	False	356	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False														DEAD_2	Helicase C-terminal domain			RTEL1 (NM_032957) - SPTLC3 (NM_018327) rearrangement: c.1141:RTEL1_c.118-6532:SPTLC3dup | Protein Fusion: mid-exon {RTEL1:SPTLC3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005100-T01-IM6-44	P-005100-T01-IM6	P-005100	Bladder Cancer	BLCA	RTEL1-TSHZ2	TSHZ2	in-frame	five_prime	62325665	30	998	True	997	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False														Helicase C-terminal domain; DEAD_2			RTEL1 (NM_032957) - TSHZ2 (NM_173485) rearrangement: c.3065-60:RTEL1_c.40+37368:TSHZ2 dup | Protein Fusion: in frame {RTEL1:TSHZ2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005100-T03-IM6-45	P-005100-T03-IM6	P-005100	Bladder Cancer	UTUC	RTEL1-TSHZ2	TSHZ2	unknown	five_prime	62325767	31	1012	False	1011	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Helicase C-terminal domain; DEAD_2			RTEL1 (NM_032957) - TSHZ2 (NM_173485) rearrangement: c.3107:RTEL1_c.41-48769:TSHZ2dup | Protein Fusion: mid-exon { RTEL1:TSHZ2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004764 8-T01-IM6-46	P-004764 8-T01-IM6	P-004764 8	Esophagogastric Cancer	ESCA	RTEL1-VRK3	VRK3	in-frame	five_prime	62325880	31	1037	True	1036	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Helicase C-terminal domain; DEAD_2			RTEL1 (NM_032957) - VRK3 (NM_016440) rearrangement: t(19;20)(q13.33;q13.33)(chr19:g.50499591::chr20:g.62325880)|Protein Fusion: in frame {RTEL1:VRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003572 3-T01-IM6-47	P-003572 3-T01-IM6	P-003572 3	Colorectal Cancer	COAD	RTEL1-YBX2	YBX2	unknown	three_prime	62292480	3	35	True	35	True	exact match	retained	1.0	False													Helicase C-terminal domain; DEAD_2			YBX2 (NM_015982) - RTEL1 (NM_032957) rearrangement: t(17;20)(p13.1;q13.33)(chr17:g.7193375::chr20:g.62292480) | Protein Fusion: mid-exon { YBX2:RTEL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002624 7-T02-IM6-48	P-002624 7-T02-IM6	P-002624 7	Ovarian Cancer	HGSOC	RTEL1-ZGPAT	ZGPAT	unknown	three_prime	62317214	16	446	False	446	True	exact match	retained	1.0	False												Helicase C-terminal domain	DEAD_2		ZGPAT (NM_032527) - RTEL1 (NM_032957) rearrangement: c.584+3556:ZGPAT_c.1409:RTEL1dup | Protein Fusion: mid-exon { ZGPAT:RTEL1 }	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003576 4-T01-IM6-49	P-003576 4-T01-IM6	P-003576 4	Soft Tissue Sarcoma	MFH	RTEL1-ZGPAT	ZGPAT	unknown	three_prime	62321528	25	744	False	744	True	exact match	lost	0.0	False														Helicase C-terminal domain; DEAD_2		ZGPAT (NM_032527) - RTEL1 (NM_032957) rearrangement: c.585-8272:ZGPAT_c.2302:RTEL1dup | Protein Fusion: mid-exon {ZGPAT: RTEL1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001439 1-T01-IM6-50	P-001439 1-T01-IM6	P-001439 1	Ovarian Cancer	HGSOC	RTEL1-ZGPAT	ZGPAT	unknown	three_prime	62326238	32	1085	False	1085	True	exact match	lost	0.0	False															Helicase C-terminal domain; DEAD_2		ZGPAT (NM_032527) - RTEL1 (NM_032957) rearrangement: c.585-11505:ZGPAT_c.3326:RTEL1dup | Protein fusion: mid-exon (ZGPAT-RTEL1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001986 2-T01-IM6-51	P-001986 2-T01-IM6	P-001986 2	Cancer of Unknown Primary	PDC	RTEL1-ZGPAT	ZGPAT	out-of-frame	five_prime	62294014	5	159	True	159	True	exact match	lost	0.0	False													Helicase C-terminal domain	DEAD_2	RTEL1 (NM_032957) - ZGPAT (NM_032527) rearrangement: c.549+34_c.585-7286del | Protein Fusion: out of frame {RTEL1:ZGPAT}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002871 6-T01-IM6-52	P-002871 6-T01-IM6	P-002871 6	Non-Small Cell Lung Cancer	LUCA	SEC23B-RTEL1	SEC23B	unknown	three_prime	62319012	17	457	False	457	True	exact match	retained	1.0	False													Helicase C-terminal domain	DEAD_2		SEC23B (NM_032986) - RTEL1 (NM_032957) rearrangement: c.-14-1432:SEC23B_c.1370:RTEL1del | Protein Fusion: mid-exon {SEC23B:RTEL1}	NA

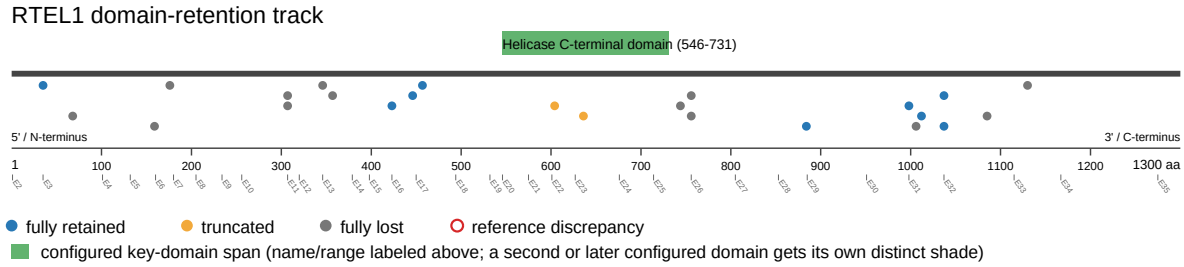
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001744 8-T01-IM6-54	P-001744 8-T01-IM6	P-001744 8	Non-Small Cell Lung Cancer	LUAD	STAU1-RTEL1	STAU1	out-of-frame	five_prime	62325877	31	1037	True	1036	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False														Helicase C-terminal domain; DEAD_2			RTEL1 (NM_032957) - STAU1 (NM_017453) rearrangement: c.3181+36:RTEL1_c.511-4612:STAU1inv | Protein Fusion: out of frame {RTEL1:STAU1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003816 4-T01-IM6-55	P-003816 4-T01-IM6	P-003816 4	Cancer of Unknown Primary	CUP	STMN3-RTEL1	STMN3	unknown	three_prime	62320417	22	604	False	604	True	exact match	disrupted	0.688172043010752 7	True	partner_lacks_domain	PF13307					7			False			DEAD_2	Helicase C-terminal domain	STMN3 (NM_015894) - RTEL1 (NM_032957) Rearrangement: c.20-3280:STMN3_c.1810:RTEL1inv | Protein Fusion: mid-exon {STMN3:RTEL1}	NA

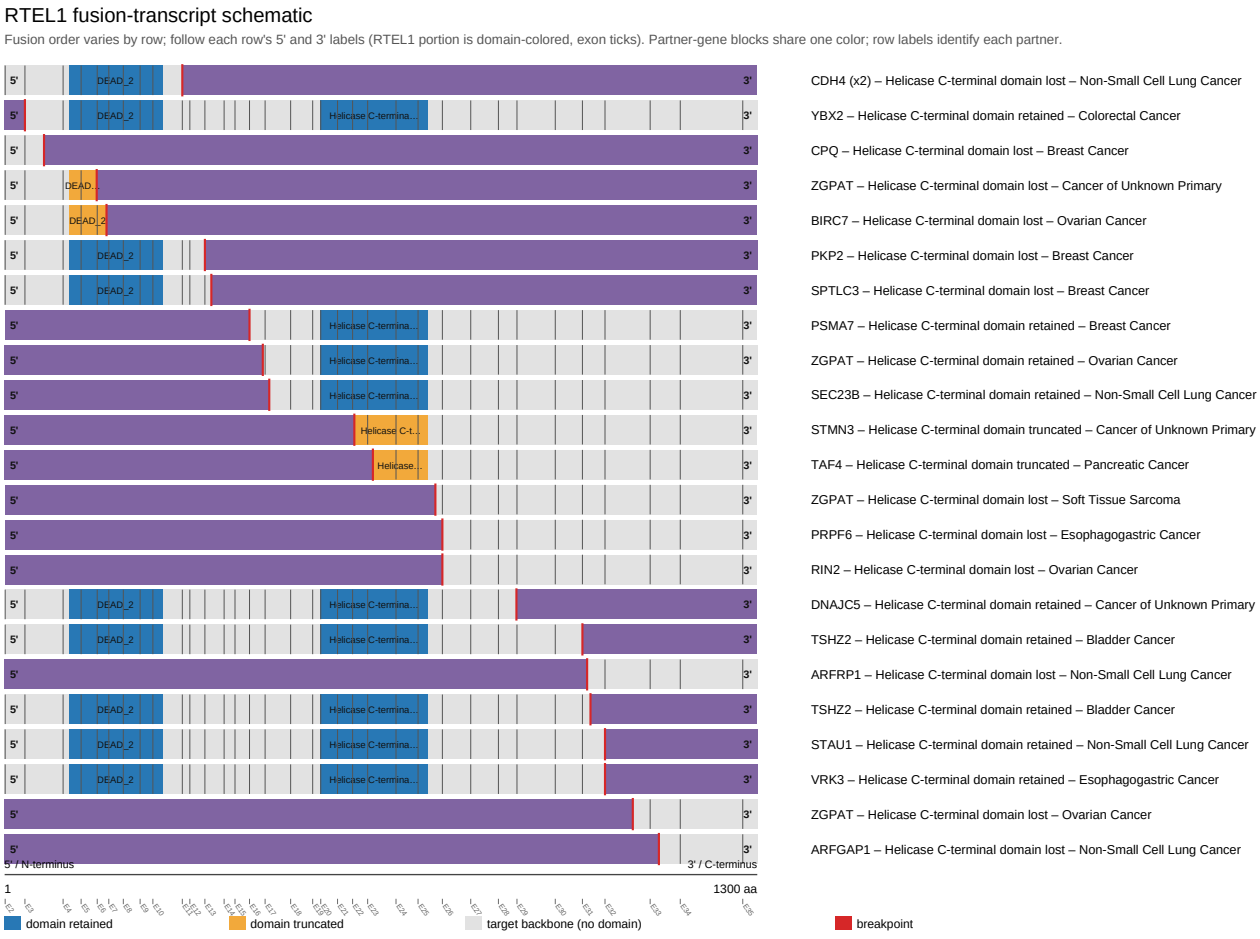
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004926 9-T01-IM6-57	P-004926 9-T01-IM6	P-004926 9	Pancreatic Cancer	PAAD	TAF4-RTEL1	TAF4	unknown	three_prime	62320884	23	636	False	636	True	exact match	disrupted	0.5161290 3225806 45	True	partner_lacks_domain	PF13307					454			False			DEAD_2	Helicase C-terminal domain	TAF4 (NM_003185) - RTEL1 (NM_032957) rearrangement: c.1360+15853:TAF4_c.1980:RTEL1inv | Protein Fusion: mid-exon {TAF4:RTEL1}	NA

Figures

domain retention outliers



fusion schematic



intragenic deletion schematic

RTEL1 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

